

TECHNICAL PRESENTATION

NAME OF THE STUDENT : J. ANISHA JENIFER

REG NO:192511004

**SIMATS**
ENGINEERING

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Sarawak Institute of Medical and Technical Sciences
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INTRODUCTION

- Air pollution is a major environmental problem that affects human health and ecosystems.
- The Air Quality Index (AQI) is used to measure pollution levels. This project presents an Air Pollution Monitoring System using C++ to analyze AQI values and classify air quality into categories like Good, Moderate, and Poor.



PROBLEM STATEMENT

- Increasing pollution due to vehicles and industries
- Lack of awareness about AQI levels
- Difficulty in understanding pollution data
- Health issues like breathing problems
- Need for simple monitoring system

Overview of Air Quality Index (AQI)

Air Quality Index (AQI) is a numerical scale used to measure air quality.

Range: 0 to 500* Lower value → Clean air
Higher value → Polluted air*Categories:

- 0-50 → Good
- 51-100 → Moderate
- 101-150 → Unhealthy
- 151-200 → Very Unhealthy
- 201-300 → Hazardous

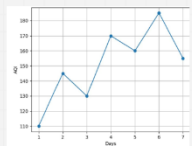


AQI Range	Category	Health Impacts
0-50	Good	Air quality is satisfactory with minimal health risk.
51-100	Moderate	Acceptable, but may affect sensitive individuals.
101-150	Unhealthy for Sensitive Groups	People with health issues may experience problems.
151-200	Unhealthy	Everyone may begin to experience health problems.
201-300	Very Unhealthy	Serious health effects for all groups.
301-500	Hazardous	Emergency conditions with major health impact.

*Higher AQI values indicate worse air quality and greater health risk.

HOW THE SYSTEM WORKS

- Takes AQI values as input
- Stores values using arrays
- Uses if-else conditions for classification
- Displays air quality category
- Provides simple output for users



TECHNOLOGY USED

1. IoT (Internet of Things)
We can use sensors to collect real-time air quality data. Devices like Arduino + gas sensors (MQ135) can be used to collect real-time AQI data using sensors.
2. Cloud Computing
Store AQI data in cloud platforms.
Access data from anywhere.
3. Machine Learning
Predict future pollution levels.
Analyze past AQI data.
Suggest preventive actions.